COMS 4170
Software Testing
Spring 2026

Chapter 3
Test Automation

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Assignment One

- Opened discussion board on Canvas
- Discussion of Handin
  - Should submit only a .pdf file. Need to show screen shots that demonstrate you have completed the assignment
  - e.g. when ask to show report provide both the summary coverage and the code view.

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Exercise One Discussion

- Branch and Line Coverage

JaCoCo Maven plug-in example with Offline Instrumentation > default

default

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
TriangleType		67%		45%	10	13	4	10	1	2	0	1
Triangle		100%		n/a	0	1	0	2	0	1	0	1
Total	16 of 76	78%	12 of 22	45%	10	14	4	12	1	3	0	2

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TriangleType.java

```
1.  //Introduction to Software Testing
2.  //Authors: Paul Ammann & Jeff Offutt
3.  //modified by M.Cohen for class exercise
4.  //Chapter 6;
5.  // See TriangleTypeTest for Junit tests
6.
7.
8.
9.
10. public class TriangleType
11. {
12.     /**
13.     * @param s1, s2, s3:  sides of the putative triangle
14.     * @return enum describing type of triangle
15.     */
16.   public static Triangle triangle (int s1, int s2, int s3)
17.   {
18.       // Reject non-positive sides
19.       if (s1 <= 0 || s2 <= 0 || s3 <= 0)
20.           return (Triangle.INVALID);
21.
22.       // Check triangle inequality
23.       if (s1+s2 <= s3 || s2+s3 <= s1 || s1+s3 <= s2)
24.           return (Triangle.INVALID);
25.
26.       // Identify equilateral triangles
27.       if ((s1 == s2) && (s2 == s3))
28.           return Triangle.EQUILATERAL;
29.
30.       // Identify isosceles triangles
31.       // original program line is below (correct version)
32.       // if ((s1 == s2) || (s2 == s3) || (s1 == s3))
33.
34.       //fault introduced below by M.Cohen
35.       if ((s1 == s2) || (s2 == s3) || (s3 == s2))
36.           return Triangle.ISOSCELES;
37.
38.       return (Triangle.SCALENE);
39.   }
40. }
```

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Covering Class Constructor

```
@Test public void createTriangle() {
TriangleType myTriangle = new TriangleType();
assertEquals(TriangleType.class, myTriangle.getClass());
}
```

```
8.
9.
10. public class TriangleType
11. {
12.     /**
13.     * @param s1, s2, s3:  sides of the putative triangle
14.     * @return enum describing type of triangle
15.     */
16.    public static Triangle triangle (int s1, int s2, int s3)
17.    {
18.       // Reject non-positive sides
19.       if (s1 <= 0 || s2 <= 0 || s3 <= 0)
20.          return (Triangle.INVALID);
21.
```

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Questions from Exercise

Change the first program check (starting on line 18):

```
// Reject non-positive sides
   //correct code - comment out
    //     if (s1 <= 0 || s2 <= 0 || s3 <= 0)

// new fault    - last condition wrong
    if (s1 <= 0 || s2 <= 0 || s3 < 0)

     return (Triangle.INVALID);
```

Can you find the fault?

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```
public class TriangleType
{
     /**
     * @param s1, s2, s3:  sides of the putative triangle
     * @return enum describing type of triangle
     */
    public static Triangle triangle (int s1, int s2, int s3)
    {
       // Reject non-positive sides
       if (s1 <= 0 || s2 <= 0 || s3 <= 0)
          return (Triangle.INVALID);

       // Check triangle inequality
       if (s1+s2 <= s3 || s2+s3 <= s1 || s1+s3 <= s2)
          return (Triangle.INVALID);
```

Test (3,3,0, INVALID)

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Running

```
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running TriangleTypeTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.029 s - in TriangleTypeTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
```

RIPR? What is happening?

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Test Automation

- From our book: "*The use of software to control the execution of tests, the comparison of actual outcomes to predicted outcomes, the setting up of test pre-conditions, and other test control and reporting functions*"
- Test generation: Another aspect of automation that can be achieved.
- We will see some automated test generation tools in this class

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Benefits of Automation

- Reduces cost of testing
- Reduces time of testing – we can test more!
- Reduces human error
- Makes regression testing easier
  - can be executed by click of a button

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Types of SE Tasks

- Revenue

  Contribute directly to the solution of a problem
- Excise

  Do not contribute directly to the solution

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Examples

- Compiling a Java class

  Excise task – necessary but not helping the problem solution
- Determining methods that are needed to define a data structure

  Revenue task –necessary and directly helps to solve the problem

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Automation

- Excise tasks are natural candidates for automation
- Software testing has more excise task than most parts of the software process
  - Maintaining test scripts
  - re-running tests
  - comparing expected results to oracle

All consume a lot of engineer time so if we can automate we benefit

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Automation

What about test generation?

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Automation

- Frees up engineer time so they can focus on the interesting work (revenue tasks)
- Allows tests to be run and re-run thousands of times and often (continuous integration)
- Helps eliminate errors of omission (e.g. failing to update new expected results)

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Software Testability

- Estimates how likely testing will reveal a fault if one exists

  i.e. how *easy or hard a software system is* to test

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Definitions

- Testability

The degree to which a system or component facilitates the establishment of test criteria and the performance of tests to determine if these criteria have been met

Determined by two common practical issues

- Observability
- Controllability

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Definitions

Software Observability

How easy it is to observe the behavior of a program in terms of its outputs, effects on the environment, and other hardware and software components.

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Definitions

- Software Controllability

How easy it is to provide a program with the needed inputs in terms of values, operations and behaviors

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Example

Embedded software

- Often does not provide human readable output, but impacts the hardware

  low observability

- Since it gets its inputs from hardware sensors (rather than a keyboard)

  low controllability

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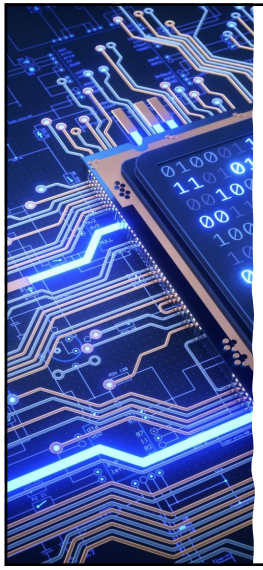

Simulation

- Used for many controllability and observability problems
- Can simulate a car crash and/or exceptional hardware values that might not be feasible with a working sensor

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Example

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Our Triangle Program

- High controllability
- High observability

```
-------------------------------------------------------
 T E S T S
-------------------------------------------------------
Running TriangleTypeTest
Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.084 sec

Results :

Tests run: 3, Failures: 0, Errors: 0, Skipped: 0

[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.863 s
[INFO] Finished at: 2025-02-03T23:41:22-06:00
[INFO] ------------------------------------------------------------------------
```

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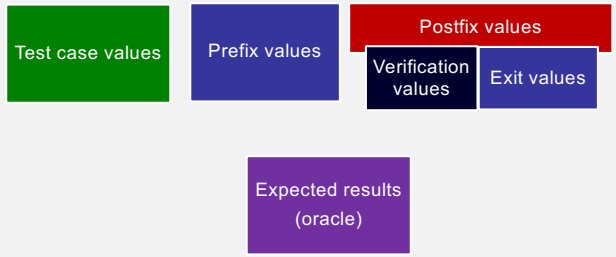
Components of a Test Case



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Components of a Test Case

Test case values

- The inputs used to complete an execution of the test
- These can be simple values, sequences, a web page, etc.

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Help with Controllability and Observability

- Prefix Values
  - Inputs necessary to put the software into an appropriate state to receive the test case values (initialization)
- Postfix Values
  - Inputs that need to be sent after the test runs (finalization)

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Example

mobile phone

- Prefix Values
  - Turning phone on
  - Opening dial pad
- Postfix Values
  - Pressing talk and end buttons

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Two Types of Postfix Values

- Verification Values
  - Necessary to see the results of the test case
- Exit values
  - Values or commands needed to terminate the program or return it to a stable state

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Test Oracle

Expected Results

- Decides whether or not a test has passed

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Test Case Definition

Composed of the *test case values, prefix values, postfix values, and expected results* necessary for completing execution and evaluation of the software under test (sometimes called the SUT)

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Test Set

A set of test cases called a test suite

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Executable Test Script

A test case/set that is prepared in a form to be executed automatically on the test software and produce a report

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RIPR Model Revisited

- Tests are designed to look for a fault in a particular location. The prefix values provide reachability (R), the test case values achieve infection (I), the postfix values achieve propagation (P) and the expected results reveal failures (R).

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Example

estimateShipping()

estimates shipping charges for preferred customers in an automated shopping cart app

Goal:

Write tests to determine if estimated charges match actual charges

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Example

- Test case values: (Infection)
  - Type of shipping (e.g. overnight)
- Prefix: (Reachability)
  - create shopping cart, add items to it, get preferred customer object and address
- Postfix: (Propagation)
  - Completing order so charges are computed
- Expected Values: (Revealability)
  - Extract actual shipping charge

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Activities we can Automate

- Prepare test inputs as executable tests
  - can use shell scripts, batch files, input files or a tool that can control the software

Ultimately the output should be a report of passing/failing along with debug info

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PART II

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Test Automation Framework

- A set of assumptions, concepts and tools that support test automation
- Should include test driver (runs a test set by executing the software on each test). Should also compare results
- Simplest test driver- main() method of a class
  - Creates instances, manipulate values, etc.

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Characteristics of Test Automation Frameworks

- Assertions to evaluate expected results
- Ability to share common test data between tests
- Test sets to easily organize and run tests
- Ability to run tests from either command line or GUI

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The JUnit Test Framework

- JUnit test scripts
  - Run as standalone Java programs (command line or from within an IDE)
- Can test an entire class, a method, interacting methods, or interactions between objects (integration)
- Tests put into Test Methods
- Test methods put into a Test Class

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Two Parts to JUnit

1. A collection of test methods
2. Methods to set up the program state (prefix values) and update after the test (postfix values)

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JUnit

- Tests written from methods in the:
  - Junit.framework.assert class
- Each test checks a condition (assertion) and reports pass/fail to the test runner

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Common Methods

assertTrue (Boolean)

simplest form (returns T/F)

assertTrue (String, Boolean)

provides information to tester if the assertion is false (failure)

fail(String)

used specify a failing message when a specific line is reached

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Example of fail()

```
@Test
public void customInterceptorAppliesWithCheckedException() {
  try {
    this.cs.throwChecked(0L);
    fail("Should have failed");
  }
  catch (RuntimeException e) {
    assertNotNull("missing original exception", e.getCause());
    assertEquals(IOException.class, e.getCause().getClass());
  }
  catch (Exception e) {
    fail("Wrong exception type " + e);
  }
}
```

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Test Fixture

- State of the test defined by the current values of key variables in the software
- Can be shared between tests
- Objects used in fixtures are declared as instance variables and initialized in a @Before method and reset or deallocated in an @After method

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Example

```
Public class Calc{

  static public int add (int a, int b)
  {
    return a+b
  }
}

'---------------------------------------'

import org.junit*;
import static org.junit.Assert.*;
public class CalcTest
{
    @Test public void testAdd()
    {
        assertEquals("Calc sum incorrect", 5==Calc.add(2,3));
    }
}
```

Expected output

Test values

Printed if assert fails

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```
public static <T extends Comparable<? super T>> T min (List<? extends T> list)
    {
      if (list.size() == 0)
      {
        throw new IllegalArgumentException ("Min.min");
      }
      Iterator<? extends T> itr = list.iterator();
      T result = itr.next();

      if (result == null) throw new NullPointerException ("Min.min");

      while (itr.hasNext())
      {  // throws NPE, CCE as needed
        T comp = itr.next();
        if (comp.compareTo (result) < 0)
        {
          result = comp;
      } }
      return result;
}
```


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MinTest Class

- Standard imports for all JUnit classes :

```
import static org.junit.Assert.*;
import org.junit.*;
import java.util.*;
```

- Test fixture and pre-test setup method (prefix) :

```
private List<String> list;   // Test fixture

// Set up - Called before every test method.
@Before
 public void setUp()
 {
    list = new ArrayList<String>();
 }
```

- Post test teardown method (postfix) :

```
// Tear down - Called after every test method.
@After
public void tearDown()
{
   list = null;   // redundant in this example
}
```


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Min Test Cases: NullPointerException

```
@Test public void testForNullList()
{
  list = null;
  try {
     Min.min (list);
  } catch (NullPointerException e) {
     return;
  }
  fail ("NullPointerException expecte
}
```

This NullPointerException test decorates the @Test annotation with the class of the exception

```
@Test (expected = NullPointerException.class)
public void testForNullElement()
{
   list.add (null);
   list.add ("cat");
   Min.min (list);
}
```

This NullPointerException test uses the fail assertion

```
@Test (expected = NullPointerException.class)
public void testForSoloNullElement()
{
   list.add (null);
   Min.min (list);
}
```

This NullPointerException test catches an easily overlooked special case

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More Exception Test Cases for Min

```
@Test (expected = ClassCastException.class)
@SuppressWarnings ("unchecked")
public void testMutuallyIncomparable()
{
  List list = new ArrayList();
  list.add ("cat");
  list.add ("dog");
  list.add (1);
  Min.min (list);
}
```

Note that Java generics don't prevent clients from using raw types!

```
@Test (expected = IllegalArgumentException.class)
public void testEmptyList()
{
   Min.min (list);
}
```

Special case: Testing for the empty list

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s for Min

```
@Test
public void testSingleElement()
{
   list.add ("cat");
   Object obj = Min.min (list);
   assertTrue ("Single Element List", obj.equals ("cat"));
}

@Test
public void testDoubleElement()
{
   list.add ("dog");
   list.add ("cat");
   Object obj = Min.min (list);
   assertTrue ("Double Element List", obj.equals ("cat"));
}
```

Finally! A couple of "Happy Path" tests

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Summary: Seven Tests for Min

- Five tests with exceptions
  1. null list
  2. null element with multiple elements
  3. null single element
  4. incomparable types
  5. empty elements
- Two without exceptions
  6. single element
  7. two elements

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Data Driven Tests

- Same test needs to run with different inputs (e.g. our triangle program)
- Write tests once and supply data in a table
- Called data-driven testing

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```
import org.junit.*;
import org.junit.runner.RunWith;
import org.junit.runners.Parameterized;
import org.junit.runners.Parameterized.Parameters;
import static org.junit.Assert.*;
import java.util.*;

@RunWith (Parameterized.class)
public class DataDrivenCalcTest
{  public int a, b, sum;

   public DataDrivenCalcTest (int v1, int v2, int expected)
   { this.a = v1; this.b = v2; this.sum = expected; }

   @Parameters public static Collection<Object[]> parameters()
   { return Arrays.asList (new Object [][] {{1, 1, 2}, {2, 3, 5}}); }

   @Test public void additionTest()
   { assertTrue ("Addition Test", sum == Calc.add (a, b)); }
}
```

Constructor is called for each triple of values

Test 1
Test values: 1, 1
Expected: 2

Test 2
Test values: 2, 3
Expected: 5

Test method

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Running JUnit on Command Line

- This is all we need to run JUnit in an IDE (like Eclipse)
- We need a main() for command line execution …

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```
import org.junit.runner.RunWith;
import org.junit.runners.Suite;
import junit.framework.JUnit4TestAdapter;

// This section declares all of the test classes in the program.
@RunWith (Suite.class)
@Suite.SuiteClasses ({ StackTest.class })  // Add test classes here.

public class AllTests
{
   // Execution begins in main(). This test class executes a
   // test runner that tells the tester if any fail.
   public static void main (String[] args)
   {
      junit.textui.TestRunner.run (suite());
   }

   // The suite() method helps when using JUnit 3 Test Runners or Ant.
   public static junit.framework.Test suite()
   {
      return new JUnit4TestAdapter (AllTests.class);
   }
}
```


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Tests with Parameters: JUnit Theories

Suppose you want to test

$$\forall x \in X \cdot P(x) \rightarrow Q(x)$$

Means: for all values in a particular domain, X, if the precondition P holds, then the postcondition Q also holds

```
@Theory public void removeThenAddDoesNotChangeSet (
        Set<String> someSet, String str) {          // Parameters!
    assumeTrue (someSet != null)                    // Assume
    assumeTrue (someSet.contains (str)) ;           // Assume
    Set<String> copy = new HashSet<String>(someSet); // Act
    copy.remove (str);
    copy.add (str);
    assertTrue (someSet.equals (copy));             // Assert
}
```


Question: Where Do The Data Values Come From?

- Answer:
  - All combinations of values from @DataPoints annotations where assume clause is true

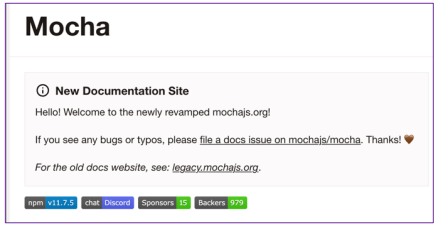
```
@DataPoints
public static String[] animals = {"ant", "bat", "cat"};

@DataPoints
public static Set[] animalSets = {
  new HashSet (Arrays.asList ("ant", "bat")),
  new HashSet (Arrays.asList ("bat", "cat", "dog", "elk")),
  new HashSet (Arrays.asList ("Snap", "Crackle", "Pop"))
};
```

Nine combinations of animalSets[i].contains (animals[j]) is false for five combinations

Summary

- The only way to make testing efficient as well as effective is to automate as much as possible
- Test frameworks provide very simple ways to automate our tests
- It is no "silver bullet" however … it does not solve the hard problem of testing:
- This is test design … the purpose of test criteria

Other (example) Unit Test Frameworks



Example PyTest

```
import time
from enum import Enum

class TriangleType(Enum):
    INVALID, EQUILATERAL, ISOSCELES, SCALENE = 0, 1, 2, 3

def classify_triangle(a, b, c):
    # Sort the sides so that a <= b <= c
    if a > b:
        tmp = a
        a = b
        b = tmp

    if a > c:
        tmp = b
        a = c
        c = tmp

    if b > c:
        tmp = b
        b = c
        c = tmp

    if a + b <= c:
        return TriangleType.INVALID
    elif a == b and b == c:
        return TriangleType.EQUILATERAL
    elif a == b or b == c:
        return TriangleType.ISOSCELES
    else:
        return TriangleType.SCALENE

if __name__ == "__main__": # pragma: no cover
    import sys
    classify_triangle(int(sys.argv[1]), int(sys.argv[2]), int(sys.argv[3]))
```

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Example PyTest

```
import time
import pytest
from triangle import TriangleType, classify_triangle


def check_classification(triangles, expected_result):
    for triangle in triangles:
        assert classify_triangle(*triangle) == expected_result


def test_invalid_triangles():
    triangles = [(1, 2, 9), (1, 9, 2), (2, 1, 9), (2, 9, 1), (9, 1, 2),
                 (9, 2, 1), (1, 1, -1), (1, -1, 1), (-1, 1, 1)]
    check_classification(triangles, TriangleType.INVALID)


def test_equilateral_triangles():
    triangles = [(1, 1, 1), (100, 100, 100), (99, 99, 99)]
    check_classification(triangles, TriangleType.EQUILATERAL)


def test_isosceles_triangles():
    triangles = [(100, 99, 99), (99, 100, 99), (99, 99, 100), (3, 3, 2), (3, 2, 3), (2, 3, 3)]

    check_classification(triangles, TriangleType.ISOCELES)


def test_scalene_triangles():
    triangles = [(5, 4, 3), (5, 3, 4), (4, 5, 3), (4, 3, 5), (3, 5, 4)]
    check_classification(triangles, TriangleType.SCALENE)


@pytest.fixture(scope="session", autouse=True)
def starter(request):
    start_time = time.time()

    def finalizer():
        print("runtime: {}".format(str(time.time() - start_time)))

    request.addfinalizer(finalizer)
```

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Results

```
mcohen> pytest -s test_triangle.py
```

```
test_triangle.py:14:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

triangles = [(1, 2, 9), (1, 9, 2), (2, 1, 9), (2, 9, 1), (9, 1, 2), (9, 2, 1), ...]
expected_result = <TriangleType.INVALID: 0>

    def check_classification(triangles, expected_result):
        for triangle in triangles:
>           assert classify_triangle(*triangle) == expected_result
E           assert <TriangleType.ISOSCELES: 2> == <TriangleType.INVALID: 0>
E            +  where <TriangleType.ISOSCELES: 2> = classify_triangle(*(2, 9, 1))

test_triangle.py:8: AssertionError
_____________________________ test_equalateral_triangles ______________________________

    def test_equalateral_triangles():
        triangles = [(1, 1, 1), (100, 100, 100), (99, 99, 99)]
>       check_classification(triangles, TriangleType.EQUALATERAL)
E       AttributeError: type object 'TriangleType' has no attribute 'EQUALATERAL'. Did
you mean: 'EQUILATERAL'?
```

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